THE MASTERMIND COACHING CLASSES

Revision Notes

Class 6 · Mathematics

Ganita Prakash — Part 1, Chapters 1 to 5 · CBSE / RBSE Syllabus

HOW TO USE THESE NOTES

Each chapter follows the textbook's own order of sections, so you can read straight through. Bold words are keywords you must be able to define. Tables compare and classify; shaded boxes hold definitions, formulas and key facts. The navy Key Takeaways box at the end of every chapter is your last-two-minutes revision.

CH.	CHAPTER TITLE	CORE THEME
1	Patterns in Mathematics	Sequences, number patterns, shape patterns, Virahānka numbers
2	Lines and Angles	Points, lines, rays, segments, angles, measuring with a protractor
3	Number Play	Number line, estimation, palindromes, Kaprekar and Collatz routines, even-odd reasoning, games
4	Data Handling and Presentation	Collecting data, tally and frequency tables, pictographs, bar graphs, reading and interpreting
5	Prime Time	Factors, multiples, primes, sieve, prime factorisation, divisibility rules, HCF and LCM

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CHAPTER 01 · MATHS (PART 1)

Patterns in Mathematics

Mathematics is the study of patterns. In this chapter you learn to spot a rule in a list of numbers or shapes, name the famous number sequences, see why they behave as they do with the help of pictures, and use the rule to predict terms far down the list.

Big Questions

- What is a sequence and how do we find its rule?
- How are square, triangular and cube numbers related to each other?
- What is the Virahānka sequence and where do such patterns appear?

1.1 Sequences and Their Rules

DEFINITION

A **sequence** is a list of numbers (or shapes) arranged in a definite order. Each member is a **term**. The **rule** tells how to get any term; three dots (…) mean the list goes on forever.

To find a rule, look at the **difference** between neighbouring terms (is it constant?), at the **ratio** (does each term double or triple?), or at the **picture** that the numbers make. Once the rule is known we can write the next terms without listing all the earlier ones.

Sequence	First terms	Rule
Counting numbers	1, 2, 3, 4, 5, 6, …	Add 1 each time; nth term = n
Odd numbers	1, 3, 5, 7, 9, 11, …	Add 2; nth odd number = 2n − 1
Even numbers	2, 4, 6, 8, 10, 12, …	Add 2; nth even number = 2n
Square numbers	1, 4, 9, 16, 25, 36, …	nth term = n × n
Triangular numbers	1, 3, 6, 10, 15, 21, …	Add 2, then 3, then 4, …; nth term = 1 + 2 + … + n
Cube numbers	1, 8, 27, 64, 125, 216, …	nth term = n × n × n
Powers of 2	1, 2, 4, 8, 16, 32, …	Double the previous term
Powers of 3	1, 3, 9, 27, 81, 243, …	Triple the previous term

REMEMBER

A rule that fits the first few terms may not be the real rule. Always test the rule on **every** given term, and look for the simplest rule that explains them all.

1.2 Seeing Patterns with Pictures

▸ Square and triangular numbers

A **square number** is the number of dots that can be arranged in a square (1, 4, 9, 16, …). A **triangular number** is the number of dots that can be stacked as a triangle with rows of 1, 2, 3, … dots. The 5th triangular number is 1 + 2 + 3 + 4 + 5 = **15**.

FORMULA

nth triangular number = 1 + 2 + 3 + … + n = **n(n + 1) ÷ 2**. Example: the 10th is 10 × 11 ÷ 2 = **55**.

▸ Three beautiful relationships

Relationship	Check	Picture reason
Sum of the first n odd numbers = n^2	1 + 3 + 5 + 7 = 16 = 4^2	Each new odd number is an L-shaped border that grows the square by one row and one column
Two consecutive triangular numbers add to a square	6 + 10 = 16, 10 + 15 = 25	Two triangles fit together to make a square of dots
Square numbers differ by odd numbers	9 − 4 = 5, 16 − 9 = 7	$(n + 1)^2 - n^2 = 2n + 1$

Worked example. Find the sum of the first 15 odd numbers. By the rule it is 15^2 = **225**. The 15th odd number itself is 2 × 15 − 1 = **29**.

Cube numbers come from stacking unit cubes into a bigger cube: 3 × 3 × 3 = 27 small cubes make a cube of side 3. Cubes grow very fast: 1, 8, 27, 64, 125, 216, 343, 512, …

1.3 The Virahānka (Fibonacci) Sequence

In ancient India, poets studied how many ways a line of n beats can be made from **short** syllables (1 beat) and **long** syllables (2 beats). The scholar **Virahānka** (and later Gopāla and Hemachandra) found that the counts follow a famous pattern.

Beats in the line	1	2	3	4	5	6
Number of ways	1	2	3	5	8	13

RULE

Every term after the first two is the **sum of the two terms just before it**: 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, … (3 + 5 = 8, 5 + 8 = 13, 21 + 34 = 55).

In the West this sequence is called the **Fibonacci sequence**. It appears in the spirals of sunflower seeds and pine cones and in the branching of plants.

1.4 Patterns in Shapes

- **Regular polygons:** 3, 4, 5, 6, … sides (triangle, square, pentagon, hexagon …): one more side each time.
- **Complete graph:** join every pair of n points by a line. Lines needed: 2 points → 1, 3 → 3, 4 → 6, 5 → 10, 6 → 15. These are the **triangular numbers** (for 6 points: 5 + 4 + 3 + 2 + 1 = 15).
- **Koch snowflake:** start with an equilateral triangle. On every line segment, replace the middle third by a ‘speed bump’ (two sides of a smaller triangle). The number of segments goes 3, 12, 48, 192, … (×4 each stage) and the shape gets a longer and longer border.

NOTE

The Koch snowflake is a **fractal**: zoom into any edge and you see the same bumpy pattern again. Fractals show how a simple repeated rule can build a very complicated figure.

1.5 Why Study Patterns?

- Patterns let us **predict** (the 100th term, the next stage of a figure).
- They explain **nature** (spirals, honeycombs, crystals) and **art** (rangoli, tiling).
- They help us **prove** general facts, such as the odd-number sum rule, instead of checking examples one by one.

Quick Revision: Likely Exam Questions

Likely question	Model answer
What is a sequence?	A list of numbers or shapes in a particular order that follows a rule, e.g. 2, 5, 8, 11, …
Why are 1, 4, 9, 16, … called square numbers?	That many dots can be arranged in a square: 1×1, 2×2, 3×3, 4×4.
Write the first six triangular numbers.	1, 3, 6, 10, 15, 21.
Find the sum of the first 9 odd numbers.	$9^2 = 81$.
Write the 7th cube number.	7 × 7 × 7 = 343.
State the rule of 3, 7, 11, 15, …. Write the next term.	Add 4 each time; next term 19.
Write the first ten Virahānka numbers.	1, 2, 3, 5, 8, 13, 21, 34, 55, 89.
How many lines join 6 points in a complete graph?	5 + 4 + 3 + 2 + 1 = 15.
Which is greater, the 7th square number or the 5th cube number?	$7^2 = 49$ and $5^3 = 125$, so the 5th cube number.
Why does the powers-of-2 sequence grow so fast?	Each term is double the one before, so it grows by multiplying, not adding.

★ KEY TAKEAWAYS

- A **sequence** follows a **rule**; find it from differences, ratios or pictures.
- **Square** numbers: n × n; **triangular**: n(n + 1) ÷ 2; **cube**: n × n × n.
- **Sum of the first n odd numbers = n^2**; two consecutive triangular numbers add to a square.
- **Virahānka (Fibonacci):** 1, 2, 3, 5, 8, 13, … each term is the sum of the two before.
- Complete-graph line counts are triangular numbers; the **Koch snowflake** is a fractal built by repeating one rule.

CHAPTER 02 · MATHS (PART 1)

Lines and Angles

Geometry starts with a point and grows into lines, rays, segments and angles. This chapter names these basic figures precisely, shows how to classify and measure angles, and uses the clock and everyday turns to build a feel for degrees.

Big Questions

- How do a line, a line segment and a ray differ?
- How are angles named, classified and measured?
- What do perpendicular and intersecting lines tell us?

2.1 Point, Line, Segment and Ray

Figure	Ends	Length	Notation
Point	none; shows a position	no size	A (a capital letter)
Line segment	two end points	can be measured	AB (segment) with length AB
Ray	one end point (starting point); goes on forever one way	cannot be measured	ray AB starts at A and passes through B
Line	no end points; goes on forever both ways	cannot be measured	line AB or a small letter l

- **Collinear points** lie on the same line; three points not on one line are non-collinear.
- **Intersecting lines** cross at exactly **one** common point. **Parallel lines** in a plane never meet.
- A line through two given points is **unique**: exactly one straight line passes through two points.

2.2 Angles and Their Parts

DEFINITION

An **angle** is formed by two rays with a **common starting point**. The rays are the **arms** and the common point is the **vertex**. In ∠PQR the vertex is Q and the arms are ray QP and ray QR (the vertex is always the middle letter).

An angle measures a **turn**. We measure it in **degrees (°)**: a full turn is 360°. A quarter turn is 90° (a **right angle**), a half turn is 180°.

Type	Measure	Example / memory aid
Acute angle	more than 0° and less than 90°	30°, 45°; ‘cute’ little angle
Right angle	exactly 90°	corner of a book; quarter turn
Obtuse angle	more than 90° and less than 180°	120°, 150°
Straight angle	exactly 180°	a straight line; half turn

Type	Measure	Example / memory aid
Reflex angle	more than 180° and less than 360°	250°, 300°
Complete angle	exactly 360°	full turn back to the start

REMEMBER

Order by size: acute < right < obtuse < straight < reflex < complete. Angles on a straight line add up to **180°**; angles around a point add up to **360°**.

2.3 Measuring Angles with a Protractor

Step 1 — Centre
Place the centre (small hole or mark) of the protractor exactly on the vertex.

Step 2 — Baseline
Turn it so that the zero line lies along one arm.

Step 3 — Scale
Start from the 0° on the arm and read along the scale that begins at 0 on that arm (inner or outer).

Step 4 — Read
Read the number where the second arm cuts the scale; extend short arms with a ruler if needed.

To **draw** an angle of 70°: draw a ray, put the protractor centre on its start, mark 70° on the correct scale, remove the protractor and join the mark to the vertex.

2.4 Angles in Daily Life: the Clock

The minute hand makes a full turn (360°) in 60 minutes, so it moves **6° per minute**. The hour hand makes a full turn in 12 hours, so it moves **30° per hour** (0.5° per minute).

Time	Angle between the hands	Type
1:00	30°	acute
3:00	90°	right
4:00	120°	obtuse
6:00	180°	straight
8:00	120° (360° − 240°)	obtuse (the smaller angle)
10:00	60°	acute

Worked example. A door opens through 60°. As a fraction of a right angle this is 60/90 = **2/3**. A wheel that turns through 3 right angles has turned 3 × 90° = **270°**, a reflex angle.

2.5 Perpendicular Lines and Estimating Angles

DEFINITION

Two lines are **perpendicular** if they meet at a right angle (90°). The perpendicular is the shortest path from a point to a line.

- Use a corner of paper or a set square to test a right angle.

- To **estimate** an angle, compare it with 90° and 45° (half of a right angle) before measuring.
- If one angle on a straight line is 110°, the other is 180° − 110° = **70°**.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Differentiate between a line, a line segment and a ray.	A line has no end points, a segment has two, a ray has one.
Define an angle and name its parts.	Two rays with a common starting point; the rays are arms and the common point is the vertex.
Name the types of angles by measure.	Acute, right, obtuse, straight, reflex, complete.
Angle turned by the minute hand in 15 minutes?	90°, a quarter turn.
Classify 30°, 90°, 150°, 180°, 220°, 360°.	Acute, right, obtuse, straight, reflex, complete.
Angle between the hands at 4 o'clock?	4 × 30° = 120°.
What are perpendicular lines?	Lines meeting at 90°.
One angle on a straight line is 110°. Find the other.	180° − 110° = 70°.
How do you read a protractor correctly?	Centre on the vertex, zero line along one arm, read the scale that starts at 0 on that arm.
How many points do two intersecting lines have in common?	Exactly one.

★ KEY TAKEAWAYS

- **Line** (no ends), **ray** (one end), **segment** (two ends, measurable).
- An angle = two rays with a common vertex; measured in **degrees**; full turn = **360°**.
- Acute < 90°, right = 90°, obtuse 90°–180°, straight = 180°, reflex 180°–360°.
- Minute hand: **6° per minute**; hour hand: **30° per hour**.
- Angles on a straight line add to **180°**; perpendicular lines meet at **90°**.

CHAPTER 03 · MATHS (PART 1)

Number Play

Numbers are fun to play with. This chapter uses the number line to compare and estimate, then explores curious number tricks (palindromes, the 1089 trick, Kaprekar's constant, the Collatz sequence), simple reasoning about odd and even numbers, and a winning strategy for a counting game.

Big Questions

- How can the number line help us compare, round and estimate?
- What surprising patterns arise from repeated reversing, adding and subtracting?
- How can we prove statements about odd and even numbers and win a counting game?

3.1 Numbers on the Number Line and Estimation

On a number line numbers increase to the right. To mark a number, choose a **scale** (for example 1,000 to 2,000 in steps of 100) and place it by counting steps; 1,730 lies between 1,700 and 1,800, a little closer to 1,700.

DEFINITION

Estimation means finding an answer that is close enough, usually by **rounding** the numbers first. To round to the nearest 1,000, look at the hundreds digit: 5 or more rounds up, less than 5 rounds down.

Calculation	Rounded to	Estimate	Exact
4,876 + 2,123	nearest 1,000	5,000 + 2,000 = 7,000	6,999
5,48,000 − 1,97,000	nearest 1,00,000	5,00,000 − 2,00,000 = 3,00,000	3,51,000
98 + 47 (mental)	98 = 100 − 2	100 + 47 − 2	145

Mental maths trick: change a number to a nearby round one and correct afterwards. 103 − 59 = 103 − 60 + 1 = **44**.

3.2 Palindromes and Number Tricks

DEFINITION

A **palindrome** reads the same from left to right and right to left, e.g. 121, 4554, 90909.

▸ Making a palindrome

Take any number, add its reverse, and repeat until a palindrome appears. Starting with 68: 68 + 86 = 154; 154 + 451 = 605; 605 + 506 = 1111, a palindrome in three steps.

▸ The 1089 trick

Step 1

Pick a 3-digit number whose first and last digits differ by at least 2, e.g. 532.

Step 2

Subtract its reverse: 532 − 235 = 297.

Step 3
Add the reverse of the result: 297 + 792 = **1089**.

▸ Kaprekar's routine (4 digits)

Arrange the digits to make the **largest** and **smallest** numbers and subtract. Repeat. Every 4-digit number (not with all digits the same) reaches **6174** within 7 steps and then stays there.

Start	Largest − smallest	Result
3087	8730 − 0378	8352
8352	8532 − 2358	6174
6174	7641 − 1467	6174 (fixed point)

▸ The Collatz sequence

Take a number: if it is **even**, halve it; if **odd**, multiply by 3 and add 1. Repeat. Example for 6: 6 → 3 → 10 → 5 → 16 → 8 → 4 → 2 → 1 (8 steps). It is believed (but not proved) that every starting number reaches 1.

> **NOTE**
> A routine that returns to the same value is said to have a **fixed point** (6174 for Kaprekar's routine).

3.3 Reasoning about Odd and Even Numbers

Operation	Result	Example
Even + even	Even	6 + 4 = 10
Odd + odd	Even	3 + 5 = 8
Even + odd	Odd	4 + 7 = 11
Even × any number	Even	6 × 7 = 42
Odd × odd	Odd	3 × 5 = 15

Why is the sum of four consecutive numbers even? Two of them are even and two are odd; odd + odd is even, even + even is even, so the total is even (3 + 4 + 5 + 6 = 18). A sum is odd only if it contains an **odd count of odd numbers**.

3.4 Divisibility Patterns

Divisible by	Test	Example
2	last digit 0, 2, 4, 6, 8	3,478
3	digit sum divisible by 3	123 → 1 + 2 + 3 = 6
4	last two digits divisible by 4	7,284 → 84
5	last digit 0 or 5	2,365
9	digit sum divisible by 9	98,765 has digit sum 35: no
10	last digit 0	4,570

3.5 A Winning Strategy: the Stick Game

A pile of 21 sticks; two players take turns removing 1, 2 or 3 sticks; whoever takes the last stick wins. **Strategy:** the first player takes 1 stick (leaving 20, a multiple of 4), and after each move of the opponent takes enough to make the pile again a **multiple of 4** (4 minus the opponent's number). Then the opponent can never reach 0.

Quick Revision: Likely Exam Questions

Likely question	Model answer
What is a palindrome? Give two examples.	A number or word that reads the same both ways, e.g. 121 and 4554.
State the 1089 trick in one line.	Subtract a 3-digit number's reverse and add the reverse of the result to get 1089.
Apply one Kaprekar step to 4153.	5431 − 1345 = 4086.
Estimate 4,876 + 2,123.	5,000 + 2,000 = 7,000.
Is the sum of two odd numbers odd or even?	Even, e.g. 3 + 5 = 8.
What is the digit sum of 98,765? Is it divisible by 9?	35; not divisible by 9.
Why is a number ending in 0 or 5 divisible by 5?	All multiples of 5 end in 0 or 5.
Difference between the smallest and largest 3-digit palindromes?	999 − 101 = 898.
Find the Collatz steps of 6.	6, 3, 10, 5, 16, 8, 4, 2, 1 (8 steps).
How do you win the 21-stick game?	Leave the opponent a multiple of 4 after each of your moves.

★ KEY TAKEAWAYS

- Use **rounding** to estimate and the **number line** to compare numbers.
- A **palindrome** reads the same both ways; reverse-and-add often creates one.
- **1089** and **6174** are fixed results of the 1089 trick and Kaprekar's routine.
- **Odd + odd = even; even × anything = even; odd × odd = odd.**
- Divisibility tests: 2 (last digit), 3 and 9 (digit sum), 4 (last two digits), 5, 10.

CHAPTER 04 · MATHS (PART 1)

Data Handling and Presentation

Facts and figures are everywhere, but they are useful only when organised. This chapter shows how to collect data, record it with tally marks, arrange it in frequency tables and present it as pictographs and bar graphs, and how to read such displays honestly.

Big Questions

- How do we collect and organise raw data?
- How are pictographs and bar graphs drawn and read?
- How can a graph mislead, and how do we avoid it?

4.1 Collecting and Organising Data

DEFINITION

Data is a collection of facts, numbers or observations gathered to get information. **Raw data** is the data as first collected, before it is arranged.

Data can be collected by **asking** (a survey), **counting**, **measuring** or **observing**. Before collecting, decide the question, the group to ask and the way to record answers.

Step 1 — Tally
Make one stroke per observation; group them in fives (four strokes crossed by a fifth) so counting is quick.

Step 2 — Count
Convert each tally group to a number called the **frequency**.

Step 3 — Table
Write categories and frequencies in a **frequency table** and check that the total equals the number of observations.

Fruit	Tally marks	Frequency
Apple	𝍸	4
Mango	||	2
Banana	|||	3
Total		**9**

Frequency of a value = the number of times it occurs. In the data 5, 3, 5, 2, 3, 5 the frequency of 5 is 3, of 3 is 2 and of 2 is 1 (total 6).

4.2 Pictographs

A **pictograph** uses a small picture or symbol to stand for a fixed number of items (the **key**). A symbol that stands for 20 books shows 60 books as 3 symbols; a half symbol shows 10 books.

Aspect	Pictograph
Strength	Attractive; quick to read; good for small numbers
Weakness	Hard to show exact values or very large numbers; half or quarter symbols are inexact
Always give	Title, key (what one symbol means), labels

4.3 Bar Graphs

DEFINITION

A **bar graph** shows data as bars of equal width with equal gaps. The **length (height)** of each bar is proportional to the value. The **scale** says how much one unit of length stands for.

Step 1
Give the graph a title and decide what each axis shows (categories on one, numbers on the other).

Step 2
Choose a **scale** so that the largest value fits, e.g. 1 cm = 10 notebooks.

Step 3
Draw bars of the same width with equal gaps; length = value ÷ scale.

Step 4
Label both axes and write the scale.

Worked example. A shop sells 40, 55, 35 and 70 notebooks in four weeks. With the scale 1 cm = 10 notebooks the bars are **4 cm, 5.5 cm, 3.5 cm and 7 cm** long. A bar of height 6 cm with scale 1 cm = 5 books shows 6 × 5 = **30 books**.

Question on a bar graph	How to answer
Which is largest/smallest?	Compare bar lengths
Total	Add all the values
Difference	Subtract the two values
How many times as much?	Divide the larger value by the smaller

Example. Rainfall: Mon 20 mm, Tue 35 mm, Wed 10 mm. Total = **65 mm**; greatest − least = 35 − 10 = **25 mm**.

4.4 Reading Graphs Wisely

- Always read the **title**, **scale** and **labels** first.
- If one bar is **twice** as long as another, its value is twice as much (when the scale starts at zero).
- A graph with **uneven bar widths** or a **cut-off scale** makes small differences look large and can mislead.
- Good presentation (neat bars, clear colours, readable lettering) helps the reader, but the **data must remain honest**.

IMPORTANT

Data is **mis-represented** when the axis does not start at zero, when bars are of different widths, or when pictures are enlarged in both directions so that they look more than twice as large.

Quick Revision: Likely Exam Questions

Likely question	Model answer
What is data? Give an example.	Facts or numbers collected to get information, e.g. marks of students.
Draw tally marks for 13.	Two bundles of five and one bundle of three.
A pictograph symbol stands for 20. What do 4½ symbols stand for?	4.5 × 20 = 90.
Why is a scale needed in a bar graph?	It decides the bar lengths and lets large numbers fit on the paper.
State two rules for drawing a bar graph.	Bars have equal width and equal gaps; axes are labelled and the graph has a title.
Frequency of 5 in 5, 3, 5, 2, 3, 5?	3.
A bar of 6 cm with scale 1 cm = 5 books shows how many books?	30 books.
Out of 50 students, 18 like cricket, 12 football, 8 hockey, the rest badminton. How many like badminton?	50 − 38 = 12.
How can a graph mislead?	Uneven widths or a scale that does not start at zero exaggerates differences.
Why should every graph have a title?	It tells the reader what the graph is about.

★ KEY TAKEAWAYS

- **Data** is organised with **tally marks** and **frequency tables**.
- A **pictograph** uses a symbol with a **key**; a **bar graph** uses bars of equal width with a **scale**.
- Bar length = value ÷ scale; value = bar length × scale.
- Read **title, scale and labels** before answering questions.
- Uneven bars or a missing zero can **mislead**; present data honestly.

CHAPTER 05 · MATHS (PART 1)

Prime Time

Primes are the atoms of numbers: every number is built from them. This chapter revises factors and multiples, finds primes with the Sieve of Eratosthenes, writes numbers as products of primes, applies divisibility tests, and uses prime factors to get the HCF and LCM.

Big Questions

- What are factors, multiples, primes and composites?
- How can every number be written in terms of primes?
- How do we find HCF and LCM and use them in problems?

5.1 Factors and Multiples

DEFINITION

A **factor** of a number divides it exactly. A **multiple** of a number is obtained by multiplying it by 1, 2, 3, … Every number is a factor and a multiple of itself; 1 is a factor of every number.

- Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36 (nine factors).
- First multiples of 7: 7, 14, 21, 28, 35, … (infinitely many).
- Factors are **finite**, multiples are **infinite**. Every factor is ≤ the number, every multiple is ≥ the number.

5.2 Primes, Composites and Co-primes

Kind	Meaning	Examples
Prime	exactly two factors: 1 and itself	2, 3, 5, 7, 11, 13, 17, 19, 23, 29
Composite	more than two factors	4, 6, 9, 15, 91 (= 7 × 13)
Neither	1 has only one factor	1
Co-primes	two numbers whose only common factor is 1	35 and 64; 8 and 15 (they need not be prime)
Twin primes	primes that differ by 2	(3, 5), (5, 7), (11, 13), (17, 19)

IMPORTANT

2 is the only even prime (every other even number has 2 as an extra factor). **1 is neither prime nor composite.** There are **10 primes below 30** and **4 between 20 and 40: 23, 29, 31, 37**.

▸ Sieve of Eratosthenes

Step 1
Write the numbers 1 to 100 in a grid and cross out 1.

Step 2
Circle 2 and cross out all its other multiples.

Step 3
Circle the next uncrossed number (3) and cross out its multiples; repeat with 5, 7.

Step 4
Stop when the next prime squared exceeds 100; all uncrossed numbers are the 25 primes up to 100.

5.3 Prime Factorisation

DEFINITION

Writing a number as a **product of primes** is its **prime factorisation**. It is **unique** apart from the order of the factors (fundamental idea of arithmetic).

Use a **factor tree**: split the number into two factors, keep splitting composite factors, stop at primes. 84 = 2 × 42 = 2 × 2 × 21 = 2 × 2 × 3 × 7 = $\mathbf{2^2 \times 3 \times 7}$. Similarly 126 = 2 × 3^2 × 7.

5.4 Divisibility Rules

By	Rule	Example
2, 5, 10	last digit even / 0 or 5 / 0	2,340
3	sum of digits divisible by 3	123: 1+2+3 = 6 ✓
9	sum of digits divisible by 9	8,172: 18 ✓
4	last two digits divisible by 4	5,316: 16 ✓
8	last three digits divisible by 8	7,392: 392 = 8 × 49 ✓
6	divisible by both 2 and 3	5,148
11	difference between the sum of digits at odd places and at even places is 0 or a multiple of 11	2,728: (2+2) − (7+8) = −11 ✓

REMEMBER

To test divisibility by a composite number such as 12, test it for **co-prime factors** (3 and 4). Testing 2 and 6 is not enough for 12 because 2 and 6 share a factor.

5.5 HCF and LCM

DEFINITION

HCF (highest common factor) is the greatest number that divides all the given numbers. **LCM** (lowest common multiple) is the smallest number that is a multiple of all of them.

Method	HCF	LCM
Prime factorisation	multiply the **common** primes with the **smallest** powers	multiply **all** primes with the **greatest** powers

Example. 12 = 2^2 × 3 and 18 = 2 × 3^2. HCF = 2 × 3 = **6**; LCM = 2^2 × 3^2 = **36**. Check: HCF × LCM = 6 × 36 = 216 = 12 × 18 (true for two numbers). HCF of 36, 48, 60 = 12.

Word problem — HCF
Cut ropes of 48 m and 36 m into equal pieces of the **longest** length: HCF = 12 m.

Word problem — LCM
Smallest number that leaves remainder 1 on division by 2, 3, 4: LCM 12, then 12 + 1 = **13**.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Define a prime number and give three examples.	A number with exactly two factors, 1 and itself: 2, 3, 5.
Is 91 prime?	No; 91 = 7 × 13.
What are twin primes? Give a pair.	Primes differing by 2; (5, 7).
List the primes between 20 and 40.	23, 29, 31, 37.
Find the prime factorisation of 84.	2 × 2 × 3 × 7.
HCF and LCM of 12 and 18?	6 and 36.
State the divisibility rule of 11.	The difference of the sums of alternate digits is 0 or a multiple of 11.
Why is every even number greater than 2 composite?	It has 2 as a factor besides 1 and itself.
Are 35 and 64 co-prime?	Yes; their HCF is 1.
Express 20 as a sum of two primes in two ways.	3 + 17 and 7 + 13.

★ KEY TAKEAWAYS

- **Prime:** exactly two factors; **composite:** more than two; **1** is neither.
- **2** is the only even prime; there are 25 primes below 100 (found by the **sieve**).
- Every number has a **unique prime factorisation** (factor tree).
- Test divisibility by a composite number through **co-prime factors**.
- **HCF** = common primes, smallest powers; **LCM** = all primes, greatest powers; for two numbers HCF × LCM = product.